

Exercise 6 – Signed Binary Numbers (2's Complement)

Signed binary numbers can also be represented using 2's complement form.

1's Complement => flip 0 to 1 and 1 to 0

$0000_2 \Rightarrow 1111$ (1's complement of 0000_2)
 $0010_2 \Rightarrow 1101$ (1's complement of 0010_2)
 $0110_2 \Rightarrow 1001$ (1's complement of 0110_2)
 $1111_2 \Rightarrow 0000$ (1's complement of 1111_2)

2's Complement => add 1 to 1's complement

$-7 \Rightarrow$ $1001_2 \Rightarrow 0110$ (1's complement of 1001_2)
 + 1

 $+7 \Leftarrow$ **0111** (2's complement of 1001_2)

$-8 \Rightarrow$ $1000 \Rightarrow 0111$ (1's complement of 1000_2)
 + 1

 $+8 \Leftarrow$ **(0)1000** (2's complement of 1000_2)

Question: Change following **4-bit** signed numbers to negative using 2's complement.

0110_2 0100_2 0001_2 0000_2

Question: Change following **4-bit** signed numbers to positive using 2's complement.

1110_2 1100_2 1001_2 1111_2

Question: Convert following decimal numbers to their negative equivalent in binary number system (using 2's complement). Use 8-bits.

10 15 18 25